

Derivatives — group rounds

Lessons 1–12 · Quarter I · 40 min · 16 questions · 80 points · groups of 3–4

Four rounds, each timed and worth more than the last. Work on paper and hand in one sheet for the group. An answer with no method scores half.

LESSON CLOCK · 40 MINUTES

4' Form groups, read the rules	5' Round A	7' Round B	8' Round C	8' Round D	8' Answers on the board, scores
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1. Groups of 3–4. One answer sheet per group, with every member’s name on it.
2. Rounds are timed and called by the teacher. When a round is called, pens down and move on — an unfinished round scores what it has.
3. In rounds C and D any member may be asked to explain the method. If nobody can, the group scores half for that question, however correct the answer is.
4. No calculators. Leave π and surds exact.

Round A · Easy

5'2 pts each

One step each — speed round

1

Find the slope of the secant to $y = x^2$ between $x = 1$ and $x = 3$.

4

Slope = $\frac{9-1}{3-1} = \frac{8}{2}$.

2

Evaluate $\lim$ as $x \rightarrow 2$ of $3x^2 - 5$.

7

A polynomial is continuous, so substitute: $3(4) - 5$.

3

Evaluate $\lim$ as $x \rightarrow 4$ of $\frac{x^2 - 16}{x - 4}$.

8

Factorise and cancel: $\frac{(x-4)(x+4)}{x-4} = x+4 \rightarrow 8$.

4

Differentiate $y = x^8$.

$8x^7$

The power rule.

5

Differentiate $y = 5x^3 - 2x$.

$15x^2 - 2$

Term by term.

6

Differentiate $y = (3x + 1)^4$.

$12(3x + 1)^3$

Chain rule: $4(3x + 1)^3 \times 3$.

Round B · Medium

7'4 pts each

Two or three steps

7

Evaluate $\lim$ as $x \rightarrow 0$ of $\frac{\sqrt{x+9} - 3}{x}$.

$\frac{1}{6}$

Multiply above and below by the conjugate $\sqrt{x+9} + 3$. The numerator becomes $(x+9) - 9 = x$, which cancels the x below, leaving $\frac{1}{\sqrt{x+9} + 3} \rightarrow \frac{1}{6}$.

8

Use first principles to find $f'(x)$ for $f(x) = x^2 - 5x$.

$2x - 5$

$\frac{(x+h)^2 - 5(x+h) - x^2 + 5x}{h} = \frac{2xh + h^2 - 5h}{h} = 2x + h - 5$, and $h \rightarrow 0$.

9

Differentiate $y = \frac{2x-3}{x+1}$.

$\frac{5}{(x+1)^2}$

Quotient rule. The numerator is $2(x+1) - (2x-3)(1) = 5$.

10

Differentiate $y = \sqrt{x^2 + 7}$.

$\frac{x}{\sqrt{x^2 + 7}}$

Chain rule: $\frac{1}{2\sqrt{x^2 + 7}} \times 2x$; the 2 cancels.

11

Where does $y = x^3 - 12x$ have a horizontal tangent?

$x = \pm 2$

$y' = 3x^2 - 12 = 0$ gives $x^2 = 4$. Two places, not one.

Round C · Hard

8'6 pts each

Two rules at once — be ready to explain

12

Differentiate $y = x^3(2x - 5)^4$ and factorise the answer.

$x^2(2x - 5)^3(14x - 15)$

Product rule with the chain rule inside: $3x^2(2x - 5)^4 + x^3 \cdot 4(2x - 5)^3 \cdot 2$. Take out $x^2(2x - 5)^3$; what is left is $3(2x - 5) + 8x = 14x - 15$.

13

Evaluate $\lim$ as $x \rightarrow 3$ of $\frac{x^2 - x - 6}{x^2 - 9}$.

$\frac{5}{6}$

Both parts are zero at $x = 3$, so factorise: $\frac{(x-3)(x+2)}{(x-3)(x+3)} = \frac{x+2}{x+3} \rightarrow \frac{5}{6}$.

14

Find the equation of the tangent to $y = \sqrt{4x + 9}$ at $x = 4$.

$y = \frac{2}{5}x + \frac{17}{5}$

$y(4) = \sqrt{25} = 5$, so the point is $(4, 5)$. $y' = \frac{2}{\sqrt{4x + 9}}$, which at $x = 4$ is $\frac{2}{5}$. Then $y - 5 = \frac{2}{5}(x - 4)$. Check: at $x = 4$ the line gives $\frac{8}{5} + \frac{17}{5} = 5$. ✓

15 A spherical balloon is inflated so that its **surface area** grows at $24\text{ cm}^2/\text{s}$.
(a) Find the rate the radius grows when $r = 3\text{ cm}$. (b) Find the rate the volume grows at that moment.
(a) $\frac{1}{\pi}\text{ cm/s}$ (b) $36\text{ cm}^3/\text{s}$
(a) $S = 4\pi r^2$, so $\frac{dS}{dr} = 8\pi r = 24\pi$ at $r = 3$. From $\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt}$:
 $24 = 24\pi \times \frac{dr}{dt}$, giving $\frac{dr}{dt} = \frac{1}{\pi}$.
(b) $V = \frac{4}{3}\pi r^3$, so $\frac{dV}{dr} = 4\pi r^2 = 36\pi$. Then $\frac{dV}{dt} = 36\pi \times \frac{1}{\pi} = 36$. The π cancels — a good sign the rate in (a) was right. The common error is to substitute $r = 3$ before differentiating, which turns the radius into a constant.

16 (a) Use first principles to show that $f(x) = \frac{1}{x}$ has $f'(x) = -\frac{1}{x^2}$.
(b) Hence find the equation of the **normal** to $y = \frac{1}{x}$ at $x = 2$.
(a) shown (b) $y = 4x - \frac{15}{2}$
(a) $\frac{f(x+h) - f(x)}{h} = \frac{1}{h}(\frac{1}{x+h} - \frac{1}{x})$. Over a common denominator the bracket is $\frac{x - (x+h)}{x(x+h)} = \frac{-h}{x(x+h)}$, so the quotient is $\frac{-1}{x(x+h)}$. Letting $h \rightarrow 0$ gives $-\frac{1}{x^2}$.
(b) At $x = 2$: the point is $(2, \frac{1}{2})$ and the tangent gradient is $-\frac{1}{4}$. The normal gradient is the negative reciprocal, 4 , so $y - \frac{1}{2} = 4(x - 2)$. Groups that use $-\frac{1}{4}$ for the normal have confused the two lines.

ROUND	QUESTIONS	EACH	ROUND TOTAL
Round A · Easy	6	2	12
Round B · Medium	5	4	20
Round C · Hard	3	6	18
Round D · Very hard	2	15	30
Total	16		80